MILITARY SPECIFICATION SHEET

MODULES, STANDARD ELECTRONIC PROGRAM

MEMORY, READ ONLY, DIGITAL

This amendment forms a part of Military Specification MIL-M-28787/74(EC), dated 13 March 1978, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title Delete "MODULES, STANDARD ELECTRONIC PROGRAM" and substitute "MODULES, STANDARD ELECTRONIC".

FSC MISC, bottom of page Delete "FSC MISC" and substitute "FSC 5963".

1.2, Module description, Environment Delete "MIL-M-28787" and substitute "MIL-STD-1389".

PAGE 2

1.2, Module description, Input capacitance, E1 and E2. Delete "35 pF maximum" and substitute "45 pF maximum".

1.2, Module description, Propagation delay time Delete "t_{EA} - Enable/disable, C_L = 50 pF" and "t_{EA} - Enable/disable, C_L = 150 pF" and substitute "t_{EA} - Enable to data out, C_L = 50 pF" and "t_{EA} - Enable to data out, C_L = 150 pF" respectively.

1.3, Module function Delete first sentence in its entirety.

PAGE 3

2, delete and substitute the following

"2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence."

PAGE 8

3 3, Failure rate requirements, third line Delete "MIL-M-28787" and substitute "MIL-STD-1389".

3.4, Performance requirements Add to end of sentence "as specified in MIL-STD-1389".

3 5 8, Marking, second line Delete "M28787/74" and substitute "M28787/74-1".

PAGE 9

TABLE II, Test conditions C21, C22, under Parameter column· Delete (4,5 V) and substitute "(4.5 V)".

PAGE 10

TABLE II, Test conditions P1-P16, under Minimum column· Delete "4" (four places) and substitute "2" (four places).

TABLE II, Test conditions N17-N32, Maximum column Delete "50" (two places) and substitute "55" (two places).

TABLE II, Test conditions P17-P32, Minimum column. Delete "4" (four places) and substitute "2" (four places).

PAGE 11

TABLE II, Test conditions W19, W20, Maximum columns. Delete "35.0" (four places) and substitute "45.0" (four places).

PAGES 12 AND 13

4 3, delete and substitute the following

"4 3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGE 14

Add new paragraph 6.3 as follows·

"6.3 Cross reference of classification. Part number M28787/74-1 supersedes part number M28787/74. The two parts are interchangeable".

PAGE 23

FIGURE 11· Delete and substitute new figure 11 as printed on page 5 of this amendment.

PAGE 27

TABLE V, Test conditions C21 and C22, under Parameter column Delete "4,5 V" and substitute "4.5 V".

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PAGE 37

FIGURE 17 Delete and substitute new figure 17 as printed on page 6 of this amendment.

PAGE 39

FIGURE 18 Delete and substitute new figure 18 as printed on page 7 of this amendment.

PAGE 42

FIGURE 20: Delete and substitute new figure 20 as printed on page 8 of this amendment.

PAGE 54

TABLE XV, Test conditions N17 through N32, under Maximum column: Delete "50.0 (two places) and substitute "55.0" (two places)

PAGE 55

TABLE XV, Test conditions P1 through P16, under Minimum column. Delete "4.0" (two places) and substitute "2" (two places).

PAGE 56

TABLE XV, Test conditions P17 through P32, under Minimum column: Delete "4.0" (two places) and substitute "2" (two places).

PAGE 58

FIGURE 25, OUTPUT PULSE 1, right side Delete "90% E" and substitute "SEE NOTE 7"; OUTPUT PULSE 2, right side, delete "90% E" and substitute "SEE NOTE 7", add new note 7 as follows

"7. For test conditions N1-N32, t_{TLH} is measured from the 10% E to 80% E points. For test conditions P1-P32, t_{THL} is measured from the 90% E to 10% E points."

PAGES 64 AND 65

FIGURE 28: Delete and substitute new figure 28 as printed on pages 9 and 10 of this amendment.

PAGE 70

* FIGURE 30 Delete and substitute new figure 30 as printed on page 11 of this amendment.

PAGE 71

* FIGURE 31: Delete and substitute new figure 31 as printed on page 12 of this amendment.

PAGE 73

FIGURE 32. Delete and substitute new figure 32 as printed on page 13 of this amendment.

PAGE 79

TABLE XXII, Test conditions W19 and W20, under Maximum column. Delete "35.0" (two places) and substitute "45.0" (two places).

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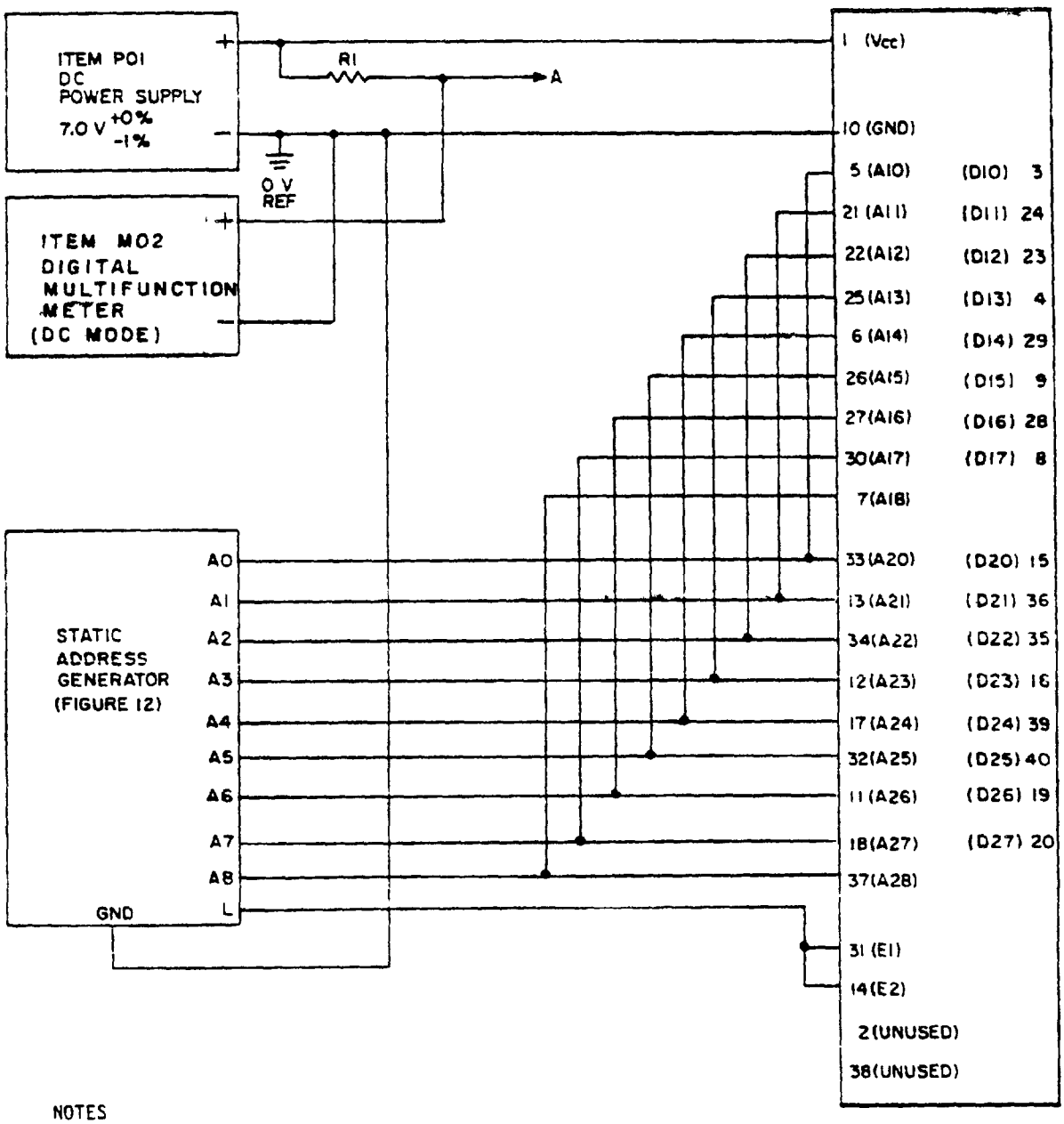
NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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DLA - ES

Preparing activity
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(Project 5963-N618)

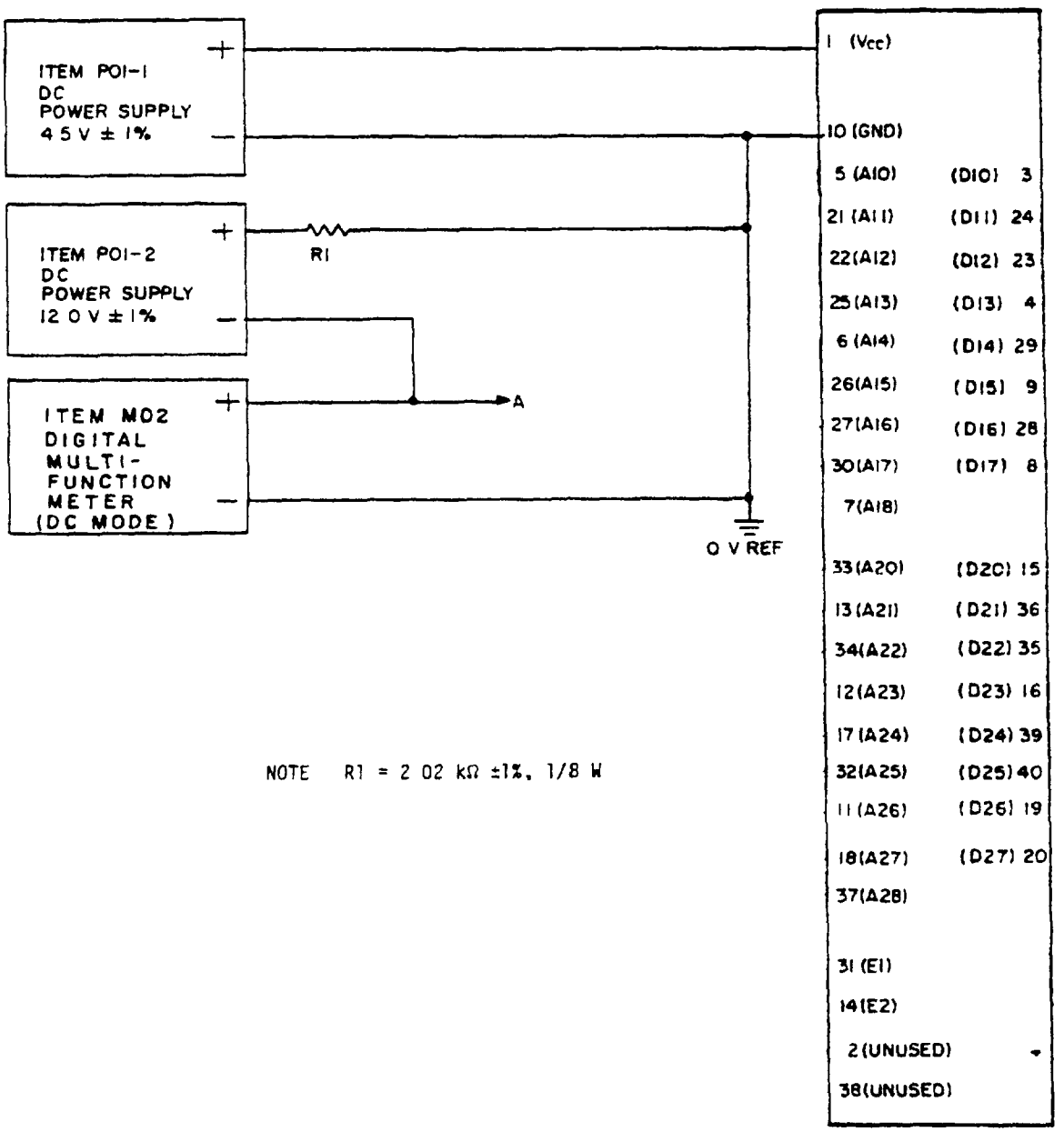
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NOTES
1 R1 = 619Ω ±1%, 1/4 W
2 For test condition B only, set the high level and low level of the static address generator to:
H = 3.5 ±0.5 V, L = 0.0 V
3 Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a low level before connecting point A. If an output is high for all addresses, omit this test for that output.

FIGURE 11 V_{out} (low-loaded) test circuit for table IV

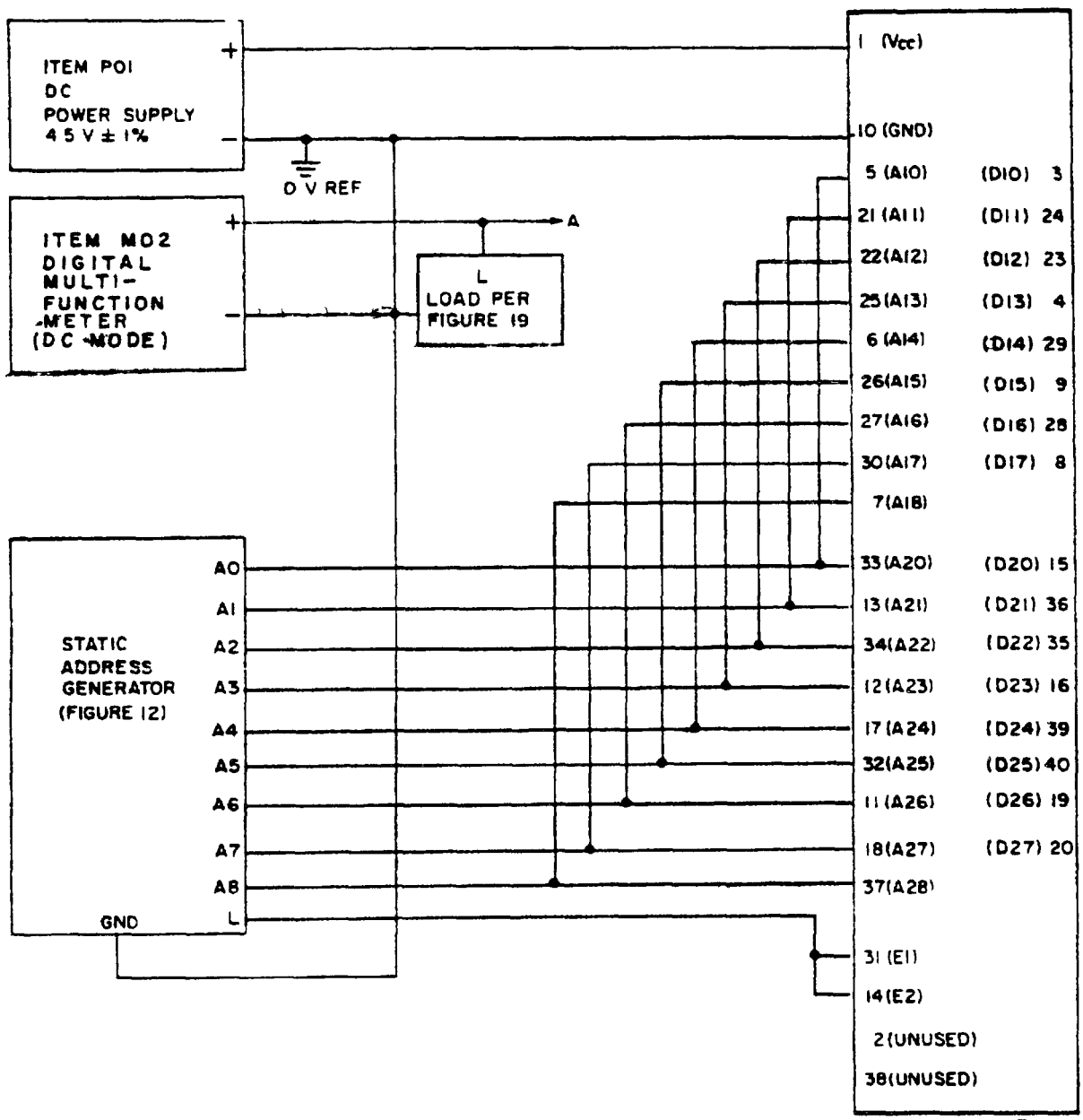
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NOTE R1 = 2 02 kΩ ±1%, 1/8 W

FIGURE 17. Input clamp diode test circuit for table IX

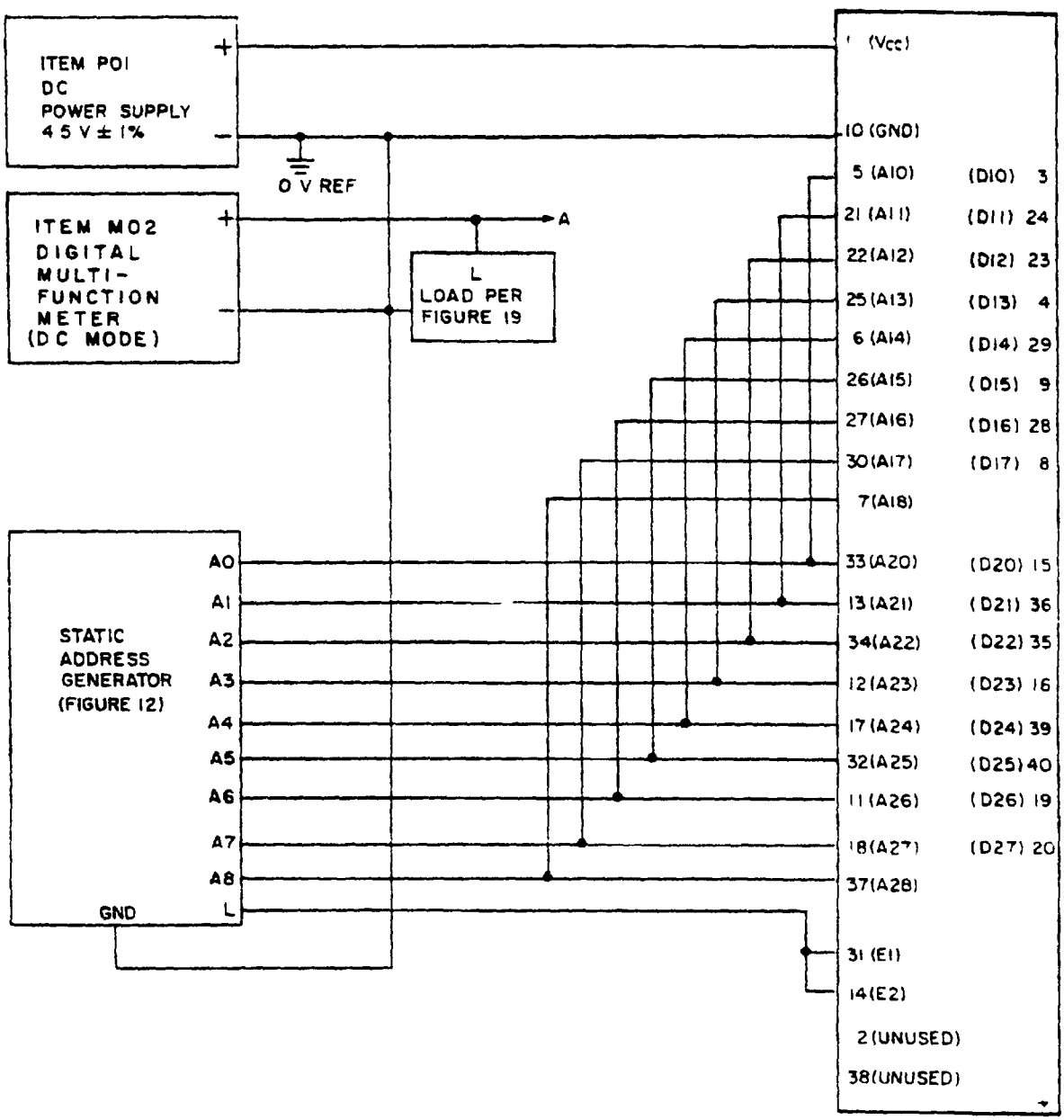
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NOTE Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a low level. If an output is high for all addresses, omit this test for that output.

FIGURE 18 Static output voltage (low-level) test circuit for table X

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NOTE Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a high level. If an output is low for all addresses, omit this test for that output.

FIGURE 20. Static output voltage (high-level) test circuit for table XI

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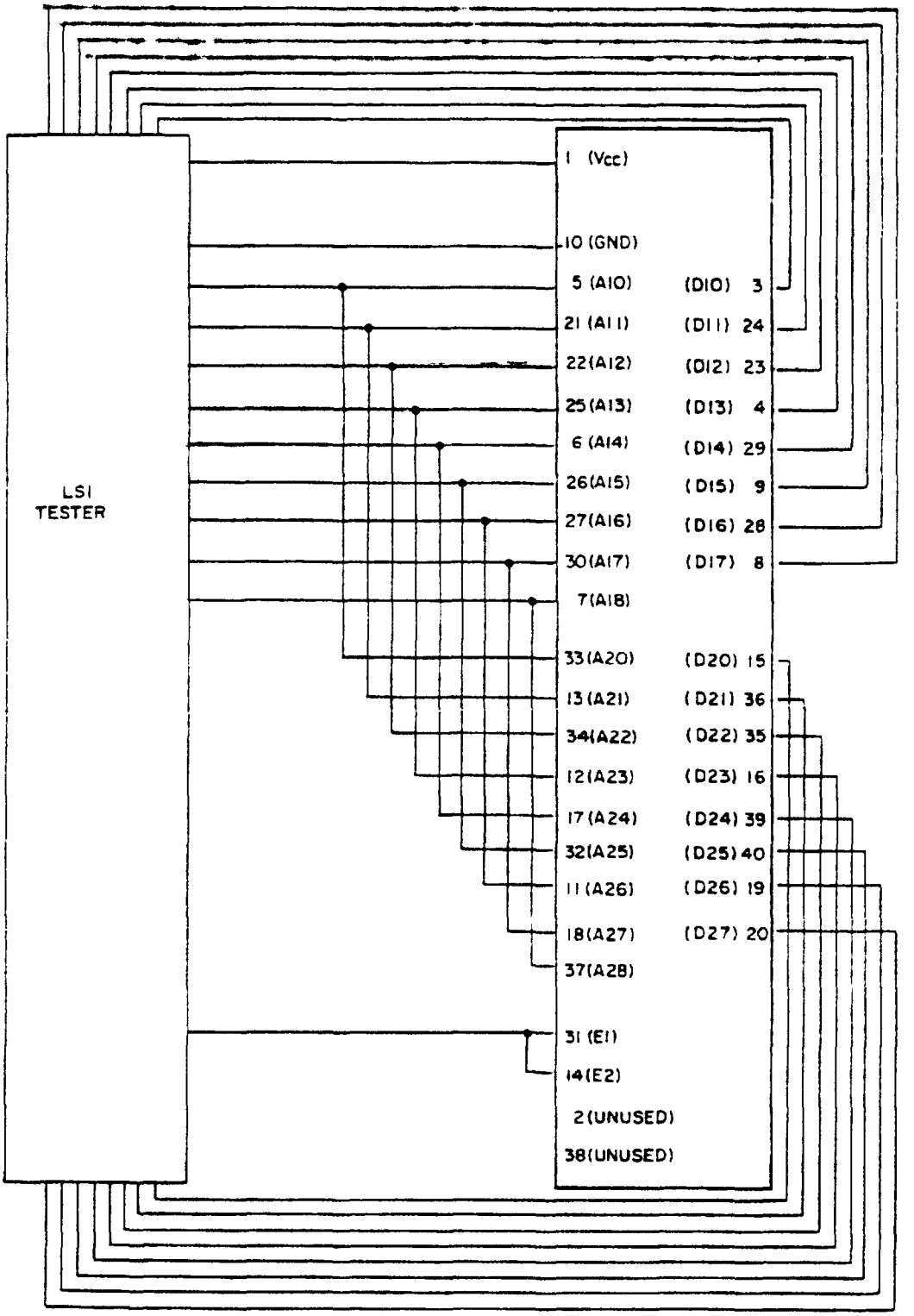


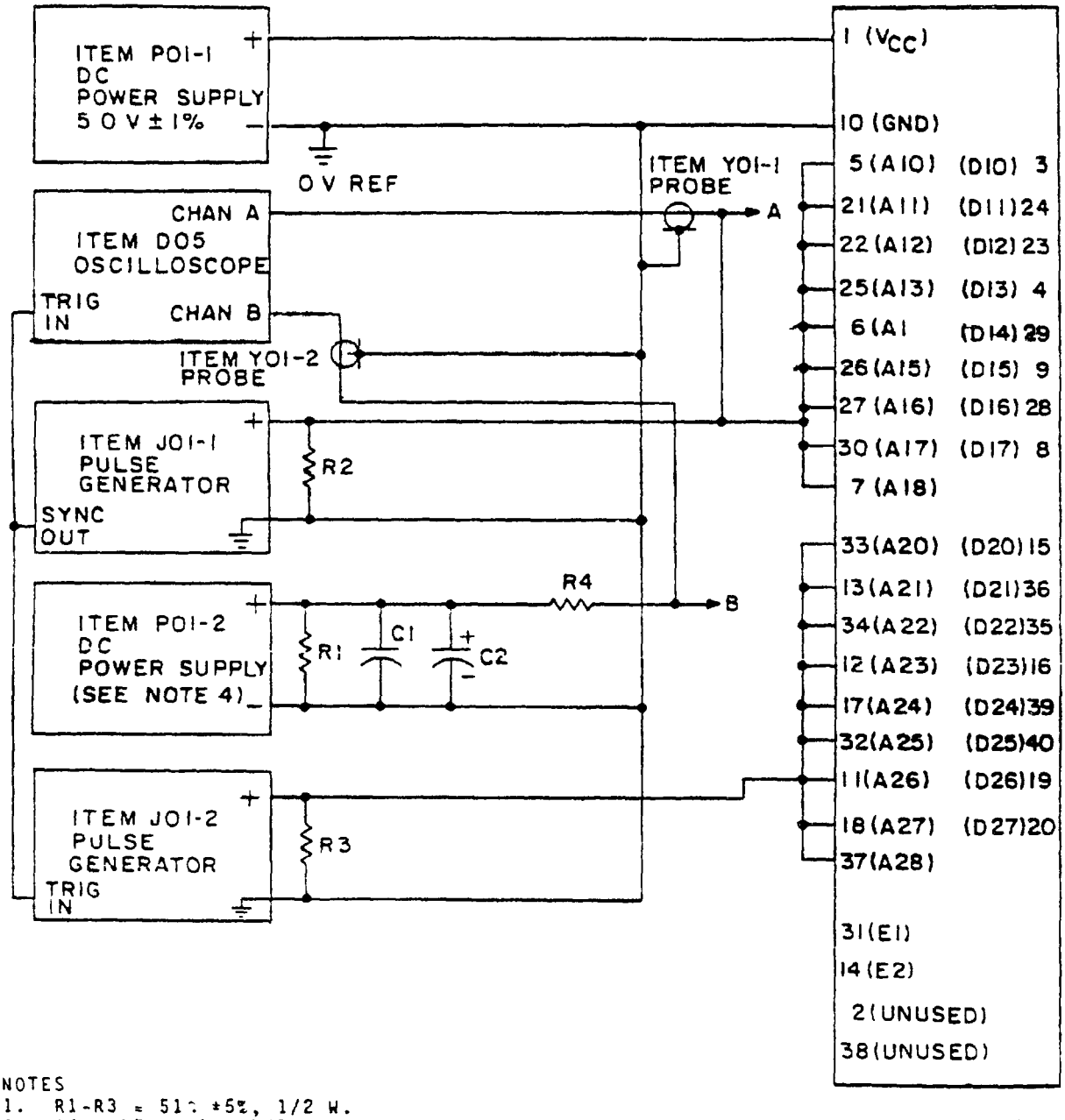
FIGURE 28 Dynamic tests test circuit for table XVII

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NOTES
1 V_{CC} = 4 5 V ±1% for test conditions R1 and R3
2 V_{CC} = 5 5 V ±1% for test conditions R2 and R4
3 Load output under test with load as specified on figure 19 with C_L = 50 pF for test conditions R1-R2
4 Load output under test with load as specified on figure 19 with C_L = 150 pF for test conditions R3-R4

FIGURE 28 Dynamic tests test circuit for table XVII - Continued

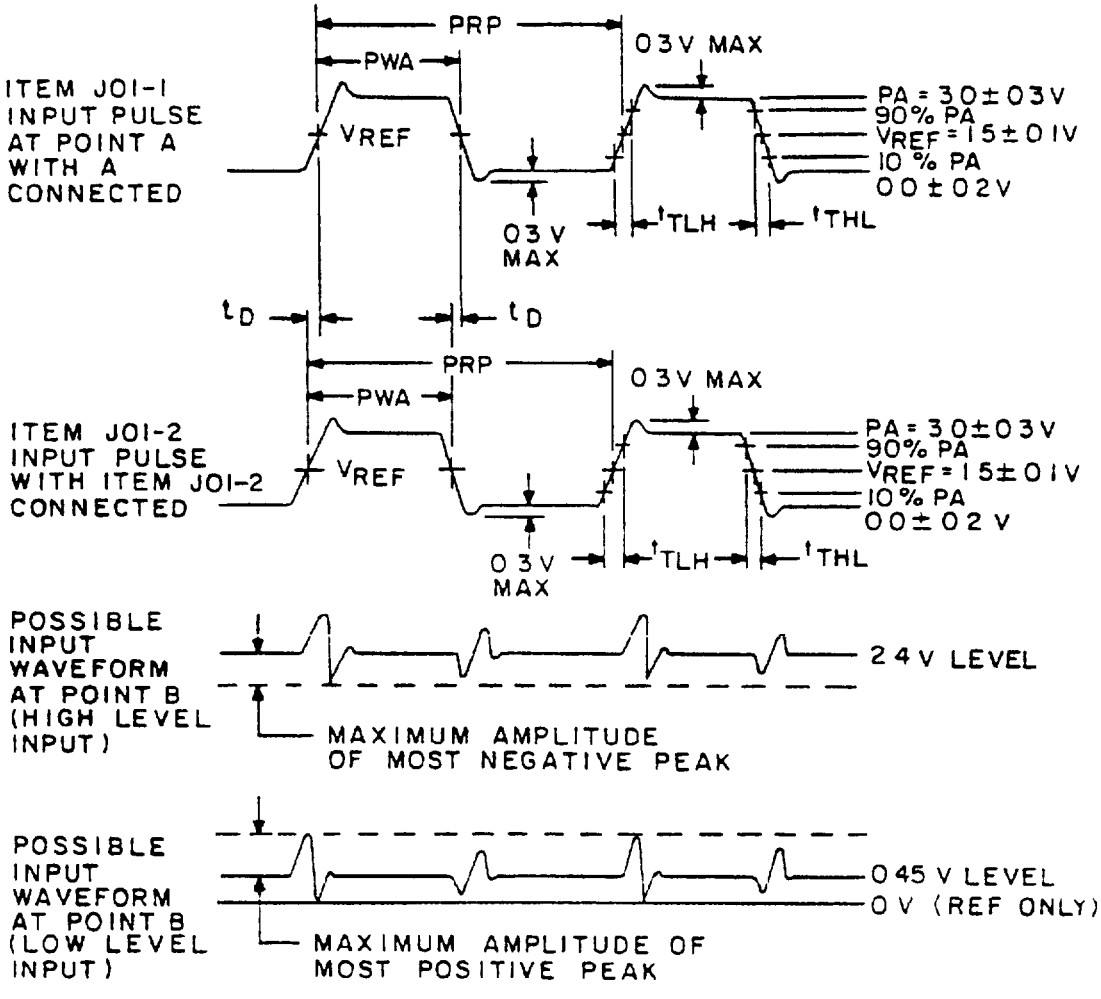
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NOTES
1. R1-R3 = 51[illegible] ±5%, 1/2 W.
2. R4 = 150[illegible] ±1%, 1/8 W.
3. Load all outputs with load as specified on figure 19.
4. For test conditions S1-S2 set item P01-2 to 0.45 V ±1%.
   For test conditions S3-S4 set item P01-2 to 2.4 V ±1%.
5. Add sufficient capacitance (C1 and C2) to reduce item P01-2 power supply noise to less than 50 mV.
6. Capacitor C2 is electrolytic.

FIGURE 30. Input crosstalk (low-level and high-level) test circuit for table XVIII.

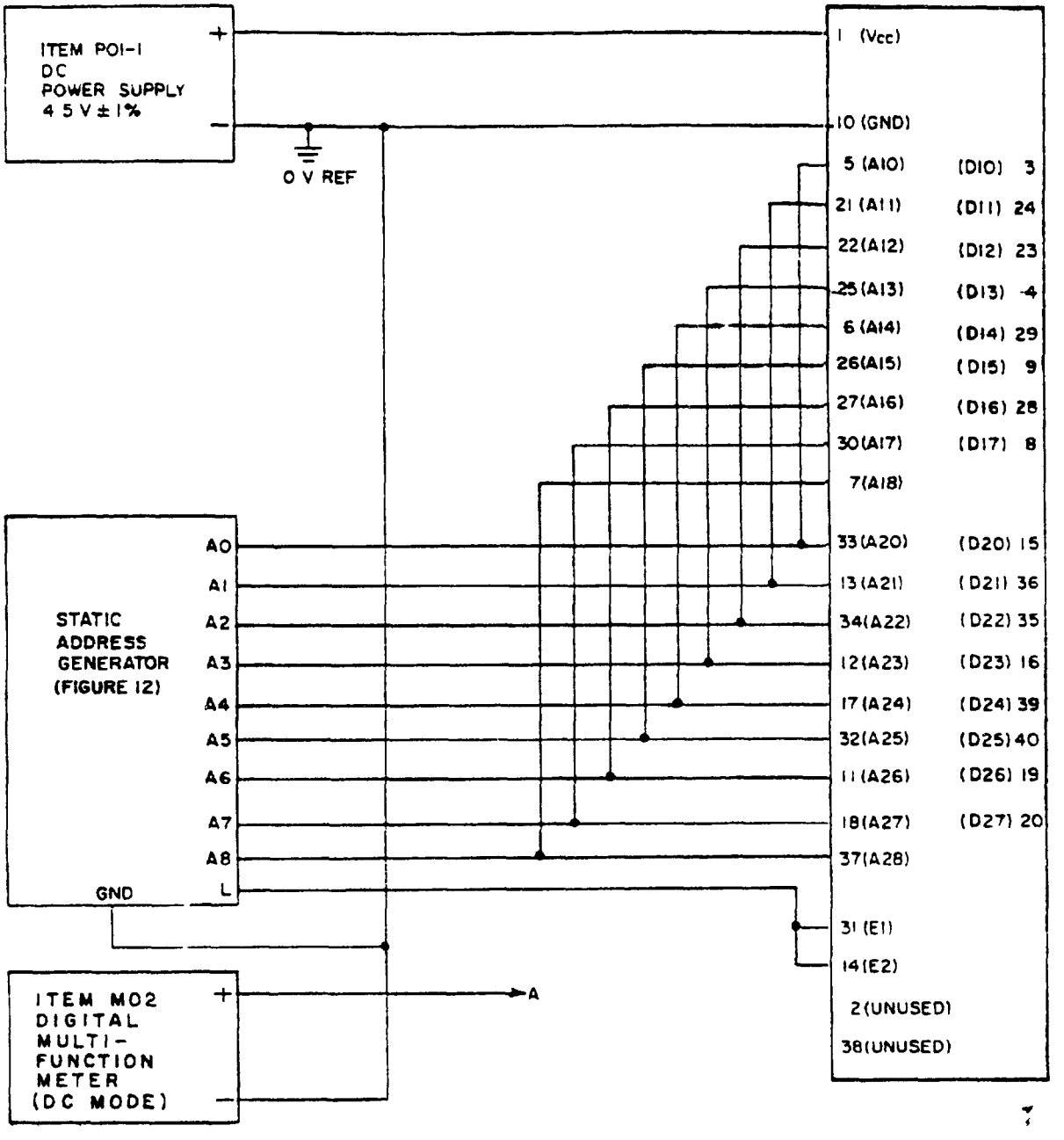
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NOTES
1. Input pulse parameters
   $t_{THL} = t_{TLH} = 20 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_D = 0 \pm 2$ ns.
2. The lead length of electrical conductors between resistors and input pins, probes Y01-1 and Y01-2 to input pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

FIGURE 31. Input crosstalk (low-level and high-level) test waveforms for table XVIII.

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NOTES
1 The static address generator is used to select, in turn, each of the 512 address combinations For each address combination the output is monitored at point A to verify the firmware pattern for the module Output high level shall be ≥ 2.4 V and low level shall be ≤ 0 5 V
2 Load the output under test with the load as specified on figure 19

FIGURE 32 Address input threshold voltage (low-level and high-level) test circuit for table XIX

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